



EDUCATION			
Program / Degree	Institution	CGPA / %	Timeline
B.Tech Engineering Design + M.Tech Computational Engineering	Indian Institute of Technology Madras	8.76 / 10.0	Aug 2023 – Present
Maharashtra Higher Secondary Certificate	Mahatma Phule Junior College	81.17%	Jun 2021 – May 2023
Central Board of Secondary Education	Jnana Prabodhini Prashala	95.4%	Jun 2015 - May 2021
- Secured All India Ranks of 2652 in JEE Advanced 2023 and 2643 in JEE Mains 2023. - Placed 2nd Globally out of 2500+ teams in Aakruti Global 2024 Design Challenge by Dassault Systemes, contributing to a top-tier design and finalist pitch on an international stage. - Patent Granted (Indian Patent No.592361) for an underactuated hyper-redundant robotic mechanism inspired by lateral undulation gait for operation in constrained environments.			
RELEVANT PROJECTS & EXPERIENCE			
Kalyani Center for Manufacturing Innovation (Bharat Forge Ltd.) <i>6-speed Gearbox Design</i>	Intern <i>May 2025 – Jul 2025</i>		
	- Translated customer needs to technical specifications using Quality Function Deployment and House of Quality Methods. - Validated a high-torque gearbox in KISSsoft, optimizing gear ratios and reducing failure risks from backlash, pitting, and scuffing. - Recommended methods to enhance drivetrain efficiency and system integration through FAST¹ analysis and Bond Graph modelling.		
Aakruti Global 2024 (Dassault Systemes) <i>1st Runner-up</i>	Team Member <i>Nov 2024</i>		
	- Ideated and modelled a safer gas stove model with in-built pan detection and heat sensing, designed in SolidWorks. - Simulated gas flow and heat transfer to validate the design and ensure reliable real-world function. - Created detailed 3D renders and animations using the 3D Excite platform to support product storytelling and evaluation. - Presented it at their Pune campus, placing 2nd among 2500+ global teams, winning a trophy and \$800 cash prize.		
Design of Mechanical Systems II (ED4060)	Course Project - Prof. Jayaganthan <i>Jan 2025 - May 2025</i>		
	- Designed a cable-driven mechanism to reduce the number of actuators required for 2D planar motion of a snake robot. - Developed a serpentine robot based on the lateral undulation gait observed in snakes; the mechanism was inspired by segmented phase sinusoidal waveforms to realize the cable-driven locomotion system for planar movement. - Tested and verified the mechanism and the control scheme in MuJoCo Physics Simulator and Patented the Mechanism (Indian Patent No.592361).		
Design of Mechanical Systems I (ED2011)	Course Project - Prof. S. Vedantam <i>Jul 2024 - Nov 2024</i>		
	- Optimized lightweight structural components in ANSYS Mechanical through FEA-driven topology and material studies, achieving ~11% material reduction while maintaining stress and safety constraints. - Performed stress, deformation, buckling, and cost analyses on an automotive component to identify an optimal steel design meeting mass, stiffness, and factor-of-safety requirements.		
Design of Fluid and Thermal Systems (ED4040)	Course Project - Prof. S. Vedantam <i>Jan 2025 - May 2025</i>		
	- Optimized an F1 front suspension wishbone using ANSYS Fluent CFD, identifying a low-lift geometry that improved aerodynamic performance while maintaining low drag. - Designed and sized a 500 W Stirling engine using MATLAB and Schmidt-cycle analysis, determining optimal operating parameters and engine geometry. - Leveraged computational fluid, thermal, and thermodynamic simulations to drive engineering design decisions across automotive and energy-conversion applications.		
Computational Methods in Design (ED5015)	Course Project - Prof. Palaniappan Ramu <i>Jan 2026 - May 2026</i>		
	- Conducted static structural FEA of a connecting rod using ANSYS Mechanical, performing mesh convergence, stress analysis, and analytical validation to accurately predict load-bearing behavior under combined axial and shear loading. - Implemented parameterized weight optimization in ANSYS APDL/Workbench, achieving ~10% mass reduction while satisfying predefined stress constraints through iterative geometry refinement.		
Foundations of Computational Fluid Dynamics (AM5630)	Course Project - Prof. Arul Prakash <i>Jan 2026 - May 2026</i>		
	- Developed MATLAB-based finite-difference solvers (FTCS, BTCS) for Parabolic PDEs and iterative solvers (Gauss–Seidel PSOR, LSOR, ADI) for elliptic PDEs. - Implemented the SMAC algorithm and Stream Function Vorticity Algorithm in MATLAB to numerically simulate incompressible channel flow. - Applied numerical methods and computational fluid dynamics techniques to solve heat transfer and fluid flow problems.		
Mechatronics Systems Design (ED5080)	Course Project - Prof. N. J. Vasa <i>Jul 2025 - Nov 2025</i>		
	Built a small 3-DoF Pick and Place Robotic Arm with color sensing to pick up and drop objects autonomously at the corresponding position based on the color.		
Advanced Geometric Computing Lab <i>Design of Indian Musical Instruments</i>	Undergraduate Research Assistant - Prof. Ramanathan <i>Feb 2026 - Jun 2026</i>		
	- Assembled and validated the first functional FDM 3D-printed Veena prototype, identifying key design deficiencies and driving iterative improvements. - Performed literature and design reviews of 3D-printed chordophones and optimized a 3D-printed electric guitar for structural performance and manufacturability. [Ongoing] Working on the redesign of the Veena, analysis for design of other Chordophones and development of a custom large-format 3D printer to advance Additive Manufacturing Research capabilities.		
Interface Lab <i>Bipedal Robot</i>	Undergraduate Research Assistant - Prof. S Ganga Prasath <i>Jul 2026 - Present</i>		
	- Designing test setup for QDD Motors to do System Identification for the Bipedal Robot. - Writing custom drivers for the QDD Motors to achieve torque control.		
POSITIONS OF RESPONSIBILITY			
CFI - CoCAS Electronics and	Contingent Captain <i>May 2025 - Apr 2026</i>		
	Spearheaded Contingent Training programs and coordinated student preparation metrics		

Robotics Contingent	Developed a repository of resources and problem statements for future Contingent Members to be trained in Mechanical Design, Machine Elements and Electronics		
	Non-Travelling Contingent Member - Eternal Problem Statement <i>Nov 2025 - Dec 2025</i>		
	Engineered a Warehouse AMR² with mecanum drive to autonomously navigate, scan shelves, and catalog items. Developed a novel Z-axis mechanism with 360-degree vision and a low-profile LiDAR chassis. Integrated sensors and motor control, using a custom PCB made with DipTrace, bridging a Teensy 4.1 and Raspberry Pi via ROS2. Spearheaded full mechatronics development, aligning custom mechanical design with complex electronic subsystems.		
CFI - CoCAS iBot Club - Team Chetak	Mechatronics Module Member <i>Jan 2024 - Jul 2024</i>		
	- Worked on the design of two competition robots through weight reduction, sizing calculations, and power/control flow design. - Helped secure a 95/100 design evaluation score in the team's inaugural competition season. - Conducted detailed competitive analysis of current and past robots, building a long-term design knowledge repository.		
	Mechatronics Lead <i>Jul 2024 - Apr 2025</i>		
	- Led the development of the two DDRobocon 2025 Robot Basketball robots, achieving a 91/100 design evaluation score. - Directed mechanism design, actuator/sensor sizing, and electrical/pneumatic/mechanical systems integration while developing physics-based ball trajectory models using Simulink and Simscape Multibody. - Applied optimization and computer vision using Simulink and Computer Vision Toolbox to calibrate simulation parameters from real-world ball motion data and oversaw prototyping and system assembly.		
Shastra - CoCAS Events and Workshops, Elecfest	Coordinator <i>Jul 2024 - Apr 2025</i>		
	- Devised Challenging problems in Digital Electronics for the various rounds of Eleckart. - Spearheaded, Brainstormed and Devised the new Elecfest event: Reverse Engineering Challenge, a challenge testing Debugging Skills in all domains of Electrical Engineering.		
EXTRA-CURRICULAR ACTIVITIES			
Marathi Mitra Mandal, IITM	Head <i>Jun 2025 - Jun 2026</i>		
	- Led Marathi Mitra Mandal as Head, managing cultural programming, operations, and stakeholder coordination to promote Marathi culture on campus. - Directed <i>Ganeshotsav 2025</i>, expanding the festival from 1 to 4 days, doubling the number of Traditional Dhols, introducing Lezim performances to the procession and managing a procession of 200+ people. Raised ₹1 lakh+ in donations for <i>Ganeshotsav 2025</i>, achieving the organization's highest fundraising total in recent years. - Managed <i>Indradhanu 2024</i> (Annual Music Event), coordinating logistics and performance execution for 40+ artists in MMM's flagship annual cultural event. - Expanded <i>Indradhanu 2025</i> (Annual Music Event) by introducing dance performances, broadening cultural representation and audience engagement. Co-authored scripts for three annual stage productions (2024–2026), contributing to creative direction and cultural storytelling. Led the culinary team for Ethnic Day 2025, coordinating planning and execution of cultural food initiatives. - Performed as a vocalist and instrumentalist in <i>Indradhanu</i> across three consecutive years (2023–2025).		
Engineering Design Academic Legislator Engineering Design Association	Founding Committee Member		
	- Co-founded the Engineering Design Association (EDA), defining the organization's vision, long-term objectives, and strategic roadmap to establish a sustainable student-led professional body. - Developed operational and fundraising frameworks, creating scalable processes for logistics management, resource allocation, and organizational growth. - Designed the association's governance structure and operating model, enabling efficient execution of initiatives and streamlined decision-making. - Authored the official EDA charter and successfully guided it through departmental approval, ensuring compliance with institutional regulations and establishing the association's formal foundation.		
Sangam - Cultural Affairs Secretary Drama Contingent	Street Play Performer <i>Inter-IIT 2024 and 2025</i>		
	Performed as an Actor and a Musician (Percussionist, Keyboardist and Flautist) in IITM’s entry to Inter-IIT Dramatics Street Play in 2024 and 2025.		
RELEVANT COURSEWORK AND SKILLS			
Geometry and Mechanics of Materials (ID6023)		Design of Mechanical Systems I (ED2011)	Design of Mechanical Systems II (ED4060)
Computational Methods in Design (ED5015)		Foundations of Computational Fluid Dynamics (AM5630)	Design of Fluid and Thermal Systems (ED4040)
Digital Signal Processing for Engineering Design (ED5017)		Mechatronic Systems Design (ED5080)	Manufacturing Processes (ED2012)
Control Systems (ED2040)		Physics of Measurement (ED2141)	Electromagnetic Compatibility for Product Design (ED5052)
Softwares: SolidWorks, Fusion 360, ANSYS Mechanical, Ansys APDL/Workbench, KISSsoft, DipTrace, MATLAB, Simulink, ROS-2, MS Excel, MuJoCo			
Programming Languages: C++, MATLAB, Python		Skills: Finite Element Analysis, System Modeling, Product Design, Prototyping, Simulation, Computational Mechanics, Additive Manufacturing	
Languages: English, Hindi, Marathi			

Abbreviations: 1. Function Analysis System Technique, 2. Autonomous Mobile Robot

